DIGITAL PSYCHROMETRIC CHART

Numerical Analysis (MATLAB) Project

Department of Mechanical Engineering
Bangladesh University of Engineering and Technology (BUET)
Dhaka

By

Md. Shahriar (1310095)

Md Jahid Hasan Sagor (1310096)

Surjo (1310097)

Eskem (1310098)

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Abstract

In this project we have developed a program that can calculate air properties for a given dry and wet bulb temperature using psychrometric calculations. As it is an important tool for engineers and engineering students, this program will help them who deal with psychrometric chart always and for applying the values for various projects or needs. Using the psychrometric chart is pesky and evasive.

Acknowledgement

We would like to express our heart-felt gratitude to our teachers who gave us the opportunity to do such projects, and encouraged us. We have to mention some of our teacher names, Dr. Mohammad Mamun, MD. Jamil Hossain, Mohd. Aminul Hoque, Satyajit Mojumder, Sourav Saha, Saif Khan Alen.

Special thanks goes to Md. Rafid Ul Islam a student of Computer Science and Engineering department, Bangladesh University of Engineering and Technology who helped us developing the android application.

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I. Introduction

This project is about digitalizing the psychrometric chart which is tedious to calculate manually. It will reduce the workload of mechanical engineers who use psychrometric chart every now and then. Our main view is to calculate all the air properties in a much more accurate way than the orthodox process. These air properties are very much important to know about the intended condition of environment.

As it is a program, it can be used in any digital device that can measure different air properties. Air Conditioners, Heaters, Industrial dryers, Washing Machines can manipulate temperature and humidity to get the required conditions. Also in cases like bigger malls, fully conditioned environments, multi storied conditioning systems, this can automatically calculate and make decisions to change the environment to a comfort or required zone.

So it will be helpful for various purposes and it can also be used industrially in a macro level.

II. Problem Specifications

Different air properties like 'relative humidity', 'humidity ratio', 'specific volume', 'enthalpy' etc. can be measured using only dry and wet bulb temperature.

Here dry and wet bulb temperature are our input variables, and the output will be several other psychrometric properties up to a certain accuracy limit. It can calculate from 0 degree dry bulb temperature to 200 degree dry bulb temperature. This range can be expanded if the file called in the code is enriched with more saturation pressure and temperature.

For example, say the dry bulb temperature and wet bulb temperature is 36 degree Celsius and 30 degree Celsius respectively. Then the program will show as following (see Fig. 2).

III. Nomenclature

`pg1`	Saturated pressure at a temperature
`patm`	Atmospheric pressure
`h`	Enthalpy
`hr`	Humidity ratio
`rh`	Relative humidity
`vol`	Specific volume

IV. Methodology

A file input is used for a set of temperature versus saturation pressure data is given. We used "Thermodynamics: An Engineering Approach" by Yunus A. Cengel for these sets of data. Later it was interpolated for more accurate values.

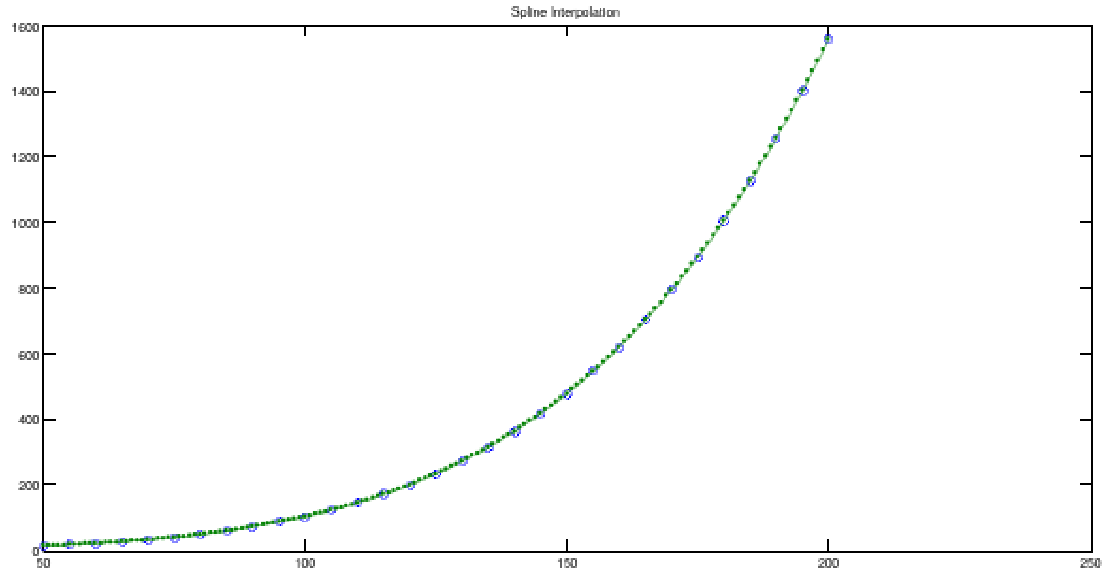

Fig. 1: Spline interpolation and rapid change of saturation pressure.

Code works on the equations stated below.

```
wg1 = 621.98*pg1./(patm-pg1);
```

It calculates a temporary humidity ratio which would only be true for equal dry and wet bulb temperature.

```
h = t1 + 2.5*wg1;
```

Then enthalpy is calculated using this equation.

```
t0 = h;
hr = (db-t1)./(t1-t0)*wg1+wg1;
```

Actual humidity ratio is calculated.

```
prs = (hr*101.325)/(621.98+hr);
rh = prs/trh*100;
```

Respective relative humidity is determined.

```
vol = rair.*(t1+273)./(patm-pg1);
```

Specific volume of air is calculated using saturated pressure and temperature.

V. Results

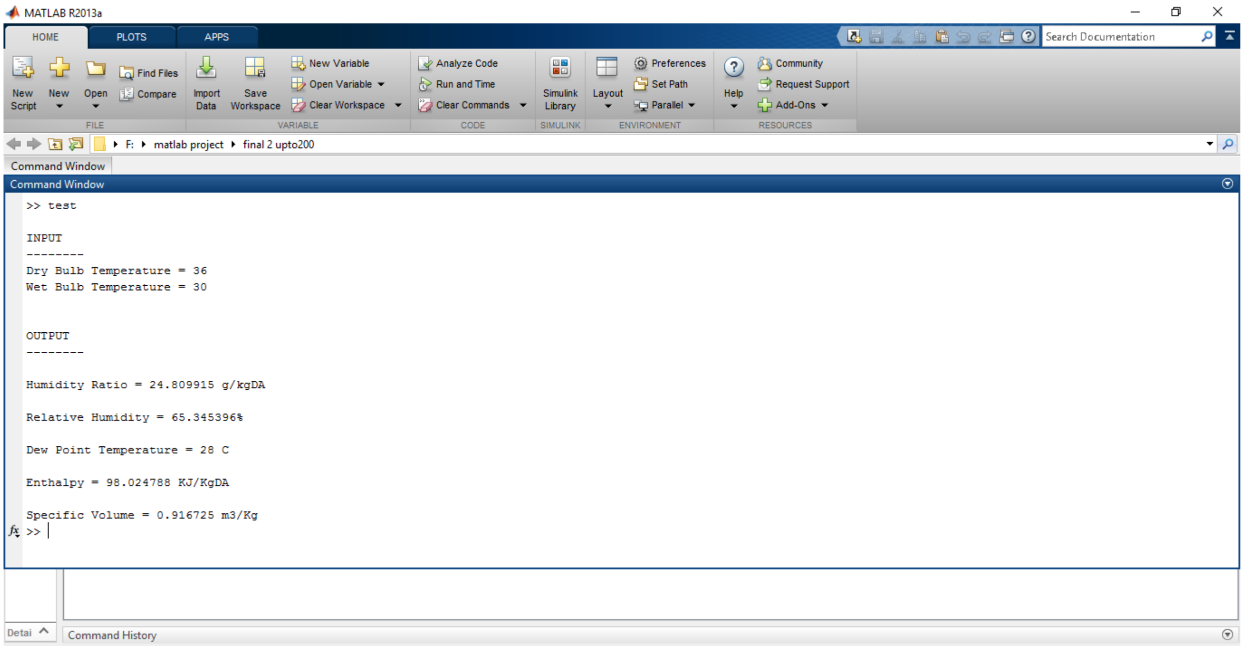

Fig. 2(a): Program input and output in the MATLAB command window.

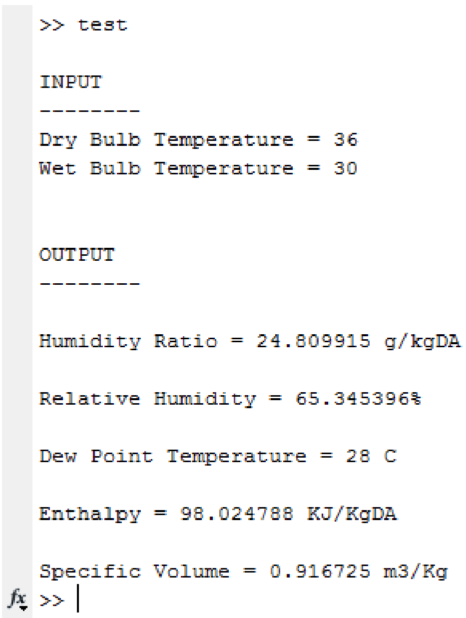

Fig. 2(b): Results.

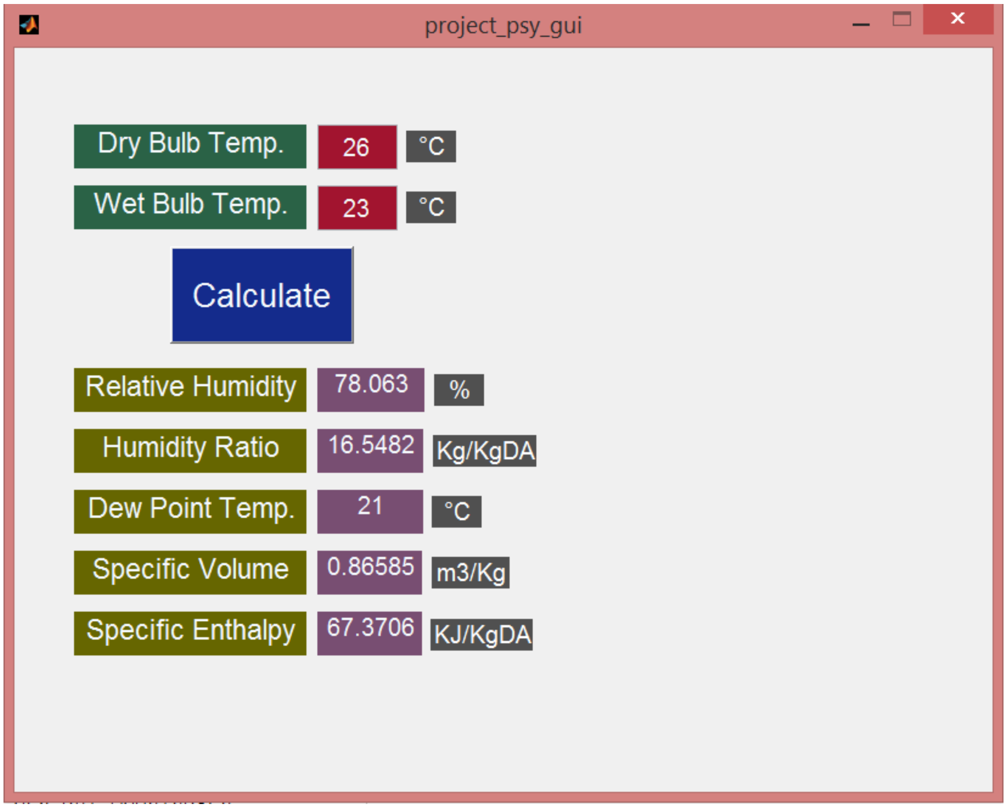

Fig. 3: Graphic user interface.

0.01	0.61165	37	6.28230	74.00	37.01033459	111.00	148.26317933	147.00	439.08409501	165.00	700.93000000
1	0.65709	38	6.63280	75.00	38.59700000	112.00	153.28163327	148.00	451.17600065	166.00	718.47600829
2	0.70599	39	7.00020	76.00	40.24050087	113.00	158.43849755	149.00	463.53327087	167.00	736.36881154
3	0.75808	40	7.38490	77.00	41.94247869	114.00	163.73690788	150.00	476.16000000	168.00	754.61361066
4	0.81355	41	7.78780	78.00	43.70460607	115.00	169.18000000	151.00	489.06037175	169.00	773.21560652
5	0.87258	42	8.20960	79.00	45.52855563	116.00	174.77086612	152.00	502.23892720	170.00	792.18000000
6	0.93536	43	8.65080	80.00	47.41600000	117.00	180.51242445	153.00	515.70029678	171.00	811.51183229
7	1.00210	44	9.11240	81.00	49.36864230	118.00	186.40754973	154.00	529.44911090	172.00	831.21550575
8	1.07300	45	9.59500	82.00	51.38830774	119.00	192.45911667	155.00	543.49000000	173.00	851.29526308
9	1.14830	46	10.09900	83.00	53.47685203	120.00	198.67000000	156.00	557.82753342	174.00	871.75534693
10	1.22820	47	10.62700	84.00	55.63613088	121.00	205.04319621	157.00	572.46603618	175.00	892.60000000
11	1.31300	48	11.17700	85.00	57.86800000	122.00	211.58218892	158.00	587.40977224	176.00	913.83394257
12	1.40280	49	11.75200	86.00	60.17432991	123.00	218.29058352	159.00	602.66300553	177.00	935.46380544
13	1.49810	50	12.35200	87.00	62.55705034	124.00	225.17198542	160.00	618.23000000	178.00	957.49669703
14	1.59900	51.00	12.97894362	88.00	65.01810581	125.00	232.23000000	161.00	634.11509457	179.00	979.93972574
15	1.70580	52.00	13.63241816	89.00	67.55944085	126.00	239.46810905	162.00	650.32292807	180.00	1002.80000000
16	1.81880	53.00	14.31352089	90.00	70.18300000	127.00	246.88929988	163.00	666.85821428	181.00	1026.08375743
17	1.93840	54.00	15.02334908	91.00	72.89078205	128.00	254.49643619	164.00	683.72566699	182.00	1049.79375248
18	2.06470	55.00	15.76300000	92.00	75.68500290	129.00	262.29238166	165.00	700.93000000	183.00	1073.93186881
19	2.19830	56.00	16.53357092	93.00	78.56793273	130.00	270.28000000	166.00	718.47600829	184.00	1098.49999010
20	2.33930	57.00	17.33615911	94.00	81.54184170	131.00	278.46236758	167.00	736.36881154	185.00	1123.50000000
21	2.48820	58.00	18.17186184	95.00	84.60900000	132.00	286.84341155	168.00	754.61361066	186.00	1148.93502772
22	2.64530	59.00	19.04177638	96.00	87.77165389	133.00	295.42727172	169.00	773.21560652	187.00	1174.81318465
23	2.81110	60.00	19.94700000	97.00	91.03195406	134.00	304.21808793	170.00	792.18000000	188.00	1201.14382772
24	2.98580	61.00	20.88864470	98.00	94.39202728	135.00	313.22000000	171.00	811.51183229	189.00	1227.93631386
25	3.16990	62.00	21.86788141	99.00	97.85400033	136.00	322.43706062	172.00	831.21550575	190.00	1255.20000000
26	3.36390	63.00	22.88589576	100.00	101.42000000	137.00	331.87297392	173.00	851.29526308	191.00	1282.94333168
27	3.56810	64.00	23.94387341	101.00	105.09227438	138.00	341.53135692	174.00	871.75534693	192.00	1311.17110891
28	3.78310	65.00	25.04300000	102.00	108.87355687	139.00	351.41582661	175.00	892.60000000	193.00	1339.88722030
29	4.00920	66.00	26.18449827	103.00	112.76670216	140.00	361.53000000	176.00	913.83394257	194.00	1369.09555446
30	4.24690	67.00	27.36973927	104.00	116.77456497	141.00	371.87754995	177.00	935.46380544	195.00	1398.80000000
31	4.49690	68.00	28.60013114	105.00	120.90000000	142.00	382.46237277	178.00	957.49669703	196.00	1429.00444554
32	4.75960	69.00	29.87708200	106.00	125.14577657	143.00	393.28842061	179.00	979.93972574	197.00	1459.71277970
33	5.03540	70.00	31.20200000	107.00	129.51432247	144.00	404.35964563	180.00	1002.80000000	198.00	1490.92889109
34	5.32510	71.00	32.57630621	108.00	134.00798007	145.00	415.68000000	181.00	1026.08375743	199.00	1522.65666832
35	5.62900	72.00	34.00147351	109.00	138.62909179	146.00	427.25345957	182.00	1049.79375248	200.00	1554.90000000
36	5.94790	73.00	35.47898770	110.00	143.38000000	147.00	439.08409501	183.00	1073.93186881		

Fig. 4: Data file.

VI. Conclusion

The outcome of this project is to introduce a digital psychrometric chart. It will be useful for digital devices that can calculate automatically the air properties and after that they can change the surrounding in such a way so that it will be turned to such a condition which is in comfort zone for human being. This calculator can calculate any value for dry bulb temperatures between 0–200 degree Celsius. If the dry bulb temperature exceeds 200 degree Celsius, then saturation pressure changes very rapidly and can cause some error. For temperature below than zero degree Celsius, we need to update the saturation pressure data (s.txt).

VII. References

1. Thermodynamics: An Engineering Approach by Yunus A. Cengel & Michael A. Boles (7th edition), Chart A5.

2.
https://www.ohio.edu/mechanical/thermo/Applied/Chapt.7_11/Psychro_chart/psychro.html

VIII. Appendix

Interpolation

```
function [ t ] = fl(db)
file = dlmread('s.txt','\t');
if floor(db)~=db
    f_db=floor(db);
    c_db=ceil(db);
    f_p=file(f_db,2);
    c_p=file(c_db,2);
    n_p = f_p + ((c_p-f_p)/(c_db-f_db))*(db-f_db);
    t=n_p;
    %disp(trh);
    %disp(c_p);
    %disp(f_p);
else
    t=file(db+1,2);
end
end
```

Main code

```
tpg = dlmread('s.txt','\t');
fprintf('\nINPUT\n--------\n');
db=input('Dry Bulb Temperature = ');
wb=input('Wet Bulb Temperature = ');

patm = 101.325;
rair = 0.287;
t1 = wb;

pg1=fl(wb);

wg1 = 621.98*pg1./(patm-pg1);
```

```
h = t1 + 2.5*wg1;
t0 = h;
hr = (db-t1)./(t1-t0)*wg1+wg1;

prs = (hr*101.325)/(621.98+hr);
trh=fl(db);

%trh = tpg(db+1,2); %pressure at dry bulb temp.
rh = prs/trh*100; %relative humidity

for i=1:200
    if prs < tpg(i,2)
        dpt=tpg(i,1)-1;
        break;
    end
end

%-----------------------
vol = rair.*(t1+273)./(patm-pg1);
%disp(vol);

tv0 = (hr/2.7475)+db;

if tv0>6.9665 & tv0<9.87215
    vol = 0.7986;
elseif tv0>9.87215 & tv0<12.7778
    vol = 0.807125;

elseif tv0>12.7778 & tv0<15.8271
    vol = 0.8160;
elseif tv0>15.8271 & tv0<18.8764
    vol = 0.8251;

elseif tv0>18.8764 & tv0<22.1132
    vol = 0.83456;
elseif tv0>22.1132 & tv0<25.3500
    vol = 0.84455;

elseif tv0>25.3500 & tv0<28.83045
    vol = 0.85495;
elseif tv0>28.83045 & tv0<32.3109
    vol = 0.86585;

elseif tv0>32.3109 & tv0<36.1072
    vol = 0.87742;
elseif tv0>36.1072 & tv0<39.9035
    vol = 0.889675;

elseif tv0>39.9035 & tv0<44.1098
    vol = 0.902775;
elseif tv0>44.1098 & tv0<48.3161
    vol = 0.916725;
end

%fprintf('\nDry Bulb Temperature = %d C\nWet Bulb Temperature = %d C\n\n',db,wb);
fprintf('\n\nOUTPUT\n--------\n');
fprintf('\nHumidity Ratio = %f g/kgDA \n\nRelative Humidity = %f%%\n\nDew Point
Temperature = %d C\n\n',hr,rh,dpt);
fprintf('Enthalpy = %f KJ/KgDA\n\nSpecific Volume = %f m3/Kg\n',h,vol);
```

Interpolating data used from books

```
x = 50:5:200;
v = [12.352 15.763 19.947 25.043 31.202 38.597 47.416 57.868 70.183 84.609 101.42
120.90 143.38 169.18 198.67 232.23 270.28 313.22 361.53 415.68 476.16 543.49 618.23
700.93 792.18 892.6 1002.8 1123.5 1255.2 1398.8 1554.9];
xq = 50:1:200;
vq2 = interp1(x,v,xq,'spline');
plot(x,v,'o',xq,vq2,':.');
xlim([50 250]);
title('Spline Interpolation');
%the values are appended to the file
```

GUI code

```
function varargout = project_psy_gui(varargin)
% PROJECT_PSY_GUI MATLAB code for project_psy_gui.fig
%      PROJECT_PSY_GUI, by itself, creates a new PROJECT_PSY_GUI or raises the
existing
%      singleton*.
%
%      TTT = PROJECT_PSY_GUI returns the handle to a new PROJECT_PSY_GUI or the handle
to
%      the existing singleton*.
%
%      PROJECT_PSY_GUI('CALLBACK',hObject,eventData,handles,...) calls the local
%      function named CALLBACK in PROJECT_PSY_GUI.M with the given input arguments.
%
%      PROJECT_PSY_GUI('Property','Value',...) creates a new PROJECT_PSY_GUI or raises
the
%      existing singleton*.  Starting from the left, property value pairs are
%      applied to the GUI before project_psy_gui_OpeningFcn gets called.  An
%      unrecognized property name or invalid value makes property application
%      stop.  All inputs are passed to project_psy_gui_OpeningFcn via varargin.
%
%      *See GUI Options on GUIDE's Tools menu.  Choose "GUI allows only one
%      instance to run (singleton)".
%
% See also: GUIDE, GUIDATA, GUIHANDLES
 % Edit the above text to modify the response to help project_psy_gui
 % Last Modified by GUIDE v2.5 04-May-2016 17:33:09
 % Begin initialization code - DO NOT EDIT
gui_Singleton = 1;
gui_State = struct('gui_Name',       mfilename, ...
                   'gui_Singleton',  gui_Singleton, ...
                   'gui_OpeningFcn', @project_psy_gui_OpeningFcn, ...
                   'gui_OutputFcn',  @project_psy_gui_OutputFcn, ...
                   'gui_LayoutFcn',  [] , ...
                   'gui_Callback',   []);
if nargin && ischar(varargin{1})
    gui_State.gui_Callback = str2func(varargin{1});
end

if nargout
    [varargout{1:nargout}] = gui_mainfcn(gui_State, varargin{:});
else
    gui_mainfcn(gui_State, varargin{:});
end
% End initialization code - DO NOT EDIT
```

```
% --- Executes just before project_psy_gui is made visible.
function project_psy_gui_OpeningFcn(hObject, eventdata, handles, varargin)
% This function has no output args, see OutputFcn.
% hObject    handle to figure
% eventdata  reserved - to be defined in a future version of MATLAB
% handles    structure with handles and user data (see GUIDATA)
% varargin   command line arguments to project_psy_gui (see VARARGIN)

% Choose default command line output for project_psy_gui
handles.output = hObject;

% Update handles structure
guidata(hObject, handles);

% UIWAIT makes project_psy_gui wait for user response (see UIRESUME)
% uiwait(handles.figure1);


% --- Outputs from this function are returned to the command line.
function varargout = project_psy_gui_OutputFcn(hObject, eventdata, handles)
% varargout  cell array for returning output args (see VARARGOUT);
% hObject    handle to figure
% eventdata  reserved - to be defined in a future version of MATLAB
% handles    structure with handles and user data (see GUIDATA)

% Get default command line output from handles structure
varargout{1} = handles.output;



function db_Callback(hObject, eventdata, handles)
% hObject    handle to db (see GCBO)
% eventdata  reserved - to be defined in a future version of MATLAB
% handles    structure with handles and user data (see GUIDATA)

% Hints: get(hObject,'String') returns contents of db as text
%        str2double(get(hObject,'String')) returns contents of db as a double


% --- Executes during object creation, after setting all properties.
function db_CreateFcn(hObject, eventdata, handles)
% hObject    handle to db (see GCBO)
% eventdata  reserved - to be defined in a future version of MATLAB
% handles    empty - handles not created until after all CreateFcns called

% Hint: edit controls usually have a white background on Windows.
%       See ISPC and COMPUTER.
if ispc && isequal(get(hObject,'BackgroundColor'),
get(0,'defaultUicontrolBackgroundColor'))
    set(hObject,'BackgroundColor','white');
end



function wb_Callback(hObject, eventdata, handles)
% hObject    handle to wb (see GCBO)
% eventdata  reserved - to be defined in a future version of MATLAB
```

```
% handles    structure with handles and user data (see GUIDATA)

% Hints: get(hObject,'String') returns contents of wb as text
%        str2double(get(hObject,'String')) returns contents of wb as a double


% --- Executes during object creation, after setting all properties.
function wb_CreateFcn(hObject, eventdata, handles)
% hObject    handle to wb (see GCBO)
% eventdata  reserved - to be defined in a future version of MATLAB
% handles    empty - handles not created until after all CreateFcns called

% Hint: edit controls usually have a white background on Windows.
%       See ISPC and COMPUTER.
if ispc && isequal(get(hObject,'BackgroundColor'),
get(0,'defaultUicontrolBackgroundColor'))
    set(hObject,'BackgroundColor','white');
end


% --- Executes on button press in pushbutton1.
function pushbutton1_Callback(hObject, eventdata, handles)
% hObject    handle to pushbutton1 (see GCBO)
% eventdata  reserved - to be defined in a future version of MATLAB
% handles    structure with handles and user data (see GUIDATA)




tpg = dlmread('t_pg.txt','\t');

db = str2double(get(handles.db,'string'));
wb = str2double(get(handles.wb,'string'));

patm = 101.325;
rair = 0.287;
t1 = wb;
pg1 = tpg(wb+1,2);
wg1 = 621.98*pg1./(patm-pg1);


h = t1 + 2.5*wg1;
t0 = h;
hr = (db-t1)./(t1-t0)*wg1+wg1;

prs = (hr*101.325)/(621.98+hr);

trh = tpg(db+1,2); %pressure at dry bulb temp.
rh = prs/trh*100; %relative humidity

for i=1:50
    if db == wb
        dpt = wb;
    elseif prs < tpg(i,2)
        dpt=tpg(i-1,1);
        disp(tpg(i,2));
        break;
    end
end
```

```
vol = rair.*(t1+273)./(patm-pg1);

tv0 = (hr/2.7475)+db;


if tv0>6.9665 & tv0<9.87215
    vol = 0.7986;
elseif tv0>9.87215 & tv0<12.7778
    vol = 0.807125;

elseif tv0>12.7778 & tv0<15.8271
    vol = 0.8160;
elseif tv0>15.8271 & tv0<18.8764
    vol = 0.8251;

elseif tv0>18.8764 & tv0<22.1132
    vol = 0.83456;
elseif tv0>22.1132 & tv0<25.3500
    vol = 0.84455;

elseif tv0>25.3500 & tv0<28.83045
    vol = 0.85495;
elseif tv0>28.83045 & tv0<32.3109
    vol = 0.86585;

elseif tv0>32.3109 & tv0<36.1072
    vol = 0.87742;
elseif tv0>36.1072 & tv0<39.9035
    vol = 0.889675;

elseif tv0>39.9035 & tv0<44.1098
    vol = 0.902775;
elseif tv0>44.1098 & tv0<48.3161
    vol = 0.916725;
end

hr = num2str(hr);
rh = num2str(rh);
vol = num2str(vol);
h = num2str(h);
dp = num2str(dpt);

set(handles.hr,'string',hr);
set(handles.rh,'string',rh);
set(handles.vol,'string',vol);
set(handles.h,'string',h);
set(handles.dp,'string',dp);
```

IX. Figures